SHANUKA BANDARA

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building full-stack web applications and applied AI solutions using MERN, Python, and modern developer tooling. Strong foundation in software engineering fundamentals, API development, database design, and disciplined version control. Demonstrated ability to deliver end-to-end projects including authentication workflows, CRUD features, RAG-based document chat, computer vision pipelines, and ML model deployment. Seeking an internship to contribute to production-minded engineering and quality-focused delivery.

EDUCATION

- **Bachelor of Science (Honors) in Computer Science** — GPA: 3.59
  South Eastern University of Sri Lanka (2022 – Present)
- **Bachelor of Information Technology (External)**
  University of Colombo School of Computing (2022 – Present)
- **Diploma in Information Technology**
  University of Colombo School of Computing (2023)

TECHNICAL SKILLS

- Languages: Python, JavaScript, Java, PHP, TypeScript (basic)
- Frontend: React.js, Tailwind CSS, Bootstrap, HTML, CSS
- Backend: Node.js, Express.js, Flask
- Databases: MongoDB, MySQL
- AI/ML: TensorFlow, Keras, OpenCV, BERT, ChromaDB
- Tools: Git, GitHub, Postman, Streamlit, CI/CD (basics)

CORE STRENGTHS

- Problem Solving & Algorithmic Thinking
- RESTful API Design & System Architecture
- Debugging, Testing & Performance Optimization
- Clean Code Practices & Version Control (Git)
- Agile Collaboration & Continuous Learning

PROJECT EXPERIENCE

Full-Stack E-Commerce Platform | MongoDB, Express.js, React, Node.js, JWT

- Developed a complete MERN-based e-commerce platform supporting seller inventory management and customer purchasing workflows
- Implemented secure JWT authentication with role-based access control for multiple user types (admin/customer)
- Designed efficient RESTful APIs and CRUD operations with optimized database interactions
- Built features such as search filtering, persistent cart functionality, and responsive UI for enhanced user experience

KnowledgeBot - AI Document Companion (RAG) | Python, Flask, Google Gemini, ChromaDB

- Built a Retrieval-Augmented Generation (RAG) chatbot enabling users to upload and query PDF documents using natural language.
- Integrated ChromaDB for vector storage and semantic search, improving response relevance.
- Implemented backend services using Flask and document parsing via PyPDF2.
- Designed a responsive frontend using Tailwind CSS for seamless interaction.

Real-Time Attendance System (Face Recognition) | Python, OpenCV, Flask, MySQL

- Engineered an attendance system using OpenCV for real-time face recognition and Flask for orchestration.
- Built a web interface for enrollment and attendance monitoring using HTML, CSS, and JavaScript.
- Integrated MySQL for secure storage of student profiles and attendance records.

AI-Powered Plant Disease Detector | Python, TensorFlow, Flask, React

- Designed a CNN-based classifier to detect and categorize 15 plant disease and healthy states from leaf images.
- Trained and optimized the model on the PlantVillage dataset using TensorFlow and Keras.
- Built a full-stack solution with a React frontend and Flask REST API providing predictions and confidence scores.

Sentimind - Product Review Analyzer | Python, BERT, Streamlit, CI/CD

- Built and deployed an NLP application that classifies review sentiment using a fine-tuned BERT model.
- Implemented batch analysis for large CSV files with progress tracking and dashboard outputs in Streamlit.
- Applied CI/CD foundations using GitHub and Streamlit Cloud for automated deployment workflows.

CERTIFICATIONS & ACHIEVEMENTS

- Full Stack Web Development (MERN) - SKYREK (PVT) LTD
- Artificial Intelligence & Machine Learning - SKYREK (PVT) LTD
- Postman API Fundamentals Student Expert - Postman
- Python for Beginners - CODL, University of Moratuwa
- TypeScript Basics - Simplilearn
- Introduction to Software Testing - Simplilearn